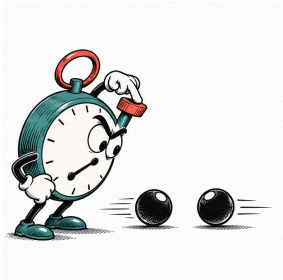
Which zeros collide first?



Tao's lattice model agrees with Haar unitary matrices on a range of low moments. Apply scalar heat to both polynomials. Stop at the first repeated root.



AN ISOLATED PAIR

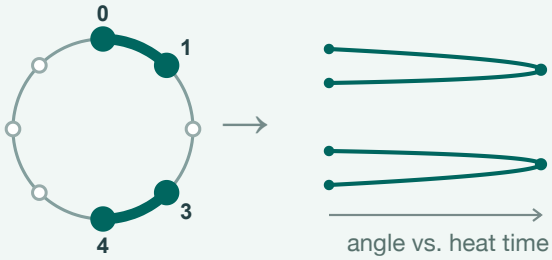
Heat lifts a shallow dip.

Near an isolated pair a gap δ looks like $x^2 - \delta^2/4$. Heat adds 2s. The two roots meet at $s = \delta^2/8$ in this quadratic model.

Four roots. The same six distances.

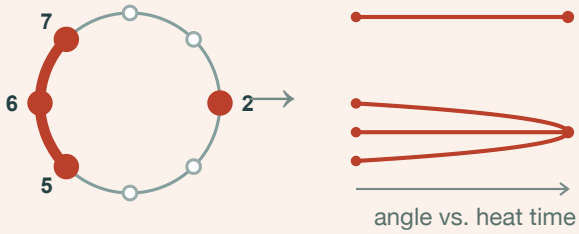
On both circles the shorter-arc distances, in grid steps, are **1, 1, 2, 3, 3, 4.**

A · Separate short edges



s ≈ 0.08863 · two double roots

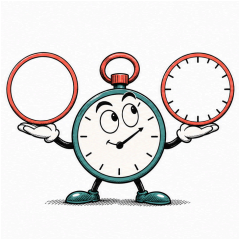
B · A shared middle root



s ≈ 0.11552 · one triple root

The missing information is which short edges share a root. That changes both the first collision time and how many roots meet.

What happens with N roots?



CUE. Eigenvalues of a Haar unitary can have a very close, isolated pair. The first collision is typically on the scale $\mathbf{N^{-8/3}}$.

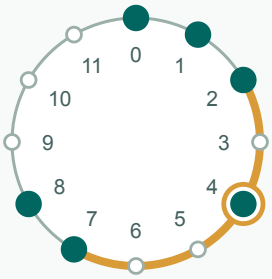
ACUE. The roots occupy N of 2N equally spaced sites. Every configuration survives at least $\boldsymbol{\pi^2/(8N^2)}$. At $s = N^{-7/3}$, the CUE collision probability tends to one. Every ACUE configuration still has distinct roots.

Why Dyson? In the Montgomery–Dyson picture, unfolded ζ zeros look locally like unitary eigenvalues. Can the primes tell us what this clock would read for ζ?

What the gap list forgets

Can we predict the motion if we also know every nearest-neighbor gap size?

At half filling, swapping occupied and empty sites preserves every pair count.



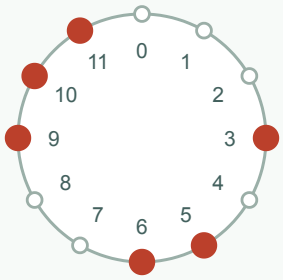
A

$T = +2$

Slope of $|p_2|^2$

+120





B

$T = -2$

Slope of $|p_2|^2$

+104

Identical pair counts at every separation. Both gap lists sort to **1, 1, 1, 2, 3, 4.**

Look at site 4 in A. Its two neighboring gaps have lengths 2 and 3. No root in B sits between gaps of those two sizes.

A cubic statistic measures the difference

Write $p_k = \sum_j e^{ik\theta_j}$. Pair counts determine $|p_k|^2 = \sum_{r,s} e^{ik(\theta_r - \theta_s)}$.

Swapping dots and holes flips both p_1 and p_2, so $T = \mathrm{Re}(p_1^2 \overline{p_2})$ changes sign.

Both start at $|p_2|^2 = 7$. Its initial slope is $16|p_2|^2 + 4T$, hence +120 and +104.

Shift a little ACUE mass between the symmetry orbits of A and B. The joint law of pair counts and sorted gaps stays fixed, while the mean slope changes. The shift also changes a cubic moment that ACUE and CUE share.

Two more ways a test can miss the difference

A window can miss a rare gap

On an unfolded CUE circle of length N, move one uniform root next to another. A fixed window of length L sees a law within $2L/N$ in total variation, even if the planted gap is tiny.

Adding Brownian noise

Run circular Dyson Brownian motion ($\beta = 2$) from CUE or ACUE. Multitime trace moments agree when total positive and negative weights are each $\le N$ across all times. A balanced witness first appears at $N + 1$.

THE QUESTION FOR ζ

Can a prime sum detect how strongly the surrounding zeros act on a close pair of zeta zeros?



Exact configurations · Window estimate · Brownian trace moments